

William Steinley

Forester / GIS Analyst · M.S. Forestry

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PROFESSIONAL SUMMARY

Forester and Geographic Information Systems (GIS) analyst with 8+ years in forestry, 2+ years of GIS analysis and LiDAR processing, an M.S. in Forestry, and California Registered Professional Forester (RPF) exam eligibility. Federal timber sale and community protection work on the Plumas National Forest. Timber sale layout and cruising in seven states. Light Detection and Ranging (LiDAR) forest structure and terrain analysis in R and Python (lidR, PDAL, GDAL). ArcGIS Pro, Field Maps, and Survey123 deployed for field data collection. M.S. thesis on fire return intervals and understory plant diversity in the East Texas Pineywoods, with a journal version as lead author.

TECHNICAL SKILLS

Forestry and Field Operations: Timber Cruising (Fixed-Plot, Variable-Radius, Strip, 100%), Silvicultural Prescription Implementation, Cable, Helicopter, and Tractor Logging Systems, Prescribed Fire and Fuels Reduction, Vegetation Management, Forest Inventory, Cruise Compilation, GPS Data Collection, Crew Supervision and Quality Control, NEPA Requirement Implementation, Timber Sale Layout

GIS and Spatial Science: ArcGIS Pro 3.x, QGIS 3.x (PyQGIS), ArcGIS Online Content Publishing and Management, Esri Field Maps, Survey123, Spatial Analysis, Remote Sensing (LiDAR, aerial imagery), LiDAR Processing (lidR, PDAL, GDAL; Canopy Height Model [CHM], Digital Terrain Model [DTM]), Canopy Cover Analysis, Geodatabase Design and Management, Geoprocessing, Cartographic Production, Coordinate Reference Systems, Slope and Terrain Modeling, Web Mapping (ArcGIS Online)

Programming and Data Management: R (lidR, terra, sf, tidyverse), Python (ArcPy, pandas, GeoPandas), Statistical Analysis, Microsoft Excel, Git and GitHub, Automated Workflow Development, Technical Writing and Documentation

WORK EXPERIENCE

Principal / Consulting Forester

Mar 2026 – Present · Self-employed, hours vary

Athena Resource Group, Gold Run, CA

- Write land management plans for private landowners. Each plan turns owner objectives into a written prescription and a treatment schedule. Part-time consulting practice with field projects in Texas.
- Perform the prescribed treatments personally: chainsaw and hand thinning, brush and invasive species control.
- Run business operations for the two-person firm: client acquisition, project scoping, and invoicing.

Forester III

Feb 2024 – Mar 2026 · Full-time, 50 hrs/wk

Jefferson Resource Company (JRC), Placerville, CA

- Supervised field crews of up to four personnel across ~8,000 acres on the Plumas National Forest Community Protection Project. Reviewed 500+ plots and caught layout and survey errors before U.S. Forest Service (USFS) deliverables went out.
- Applied National Environmental Policy Act (NEPA) requirements and silvicultural prescriptions in the field, including layout of tractor, cable and helicopter yarding units.
- Led the move from Avenza Maps Pro to ArcGIS Online, ArcGIS Pro, Esri Field Maps, and Survey123. Wrote two technical guides and the training materials that brought the crews over. Ended manual re-keying of field data and gave crews real-time cloud access.
- Automated map production and spatial data processing in Python and ArcPy, replacing hand-assembled maps with one standard output across deliverables. Validated every output against manual quality-control checks.
- Built CHM and DTM products from LiDAR point clouds in R (lidR). Checked point density and compared resampling approaches before setting the grid resolution. Produced slope and terrain derivatives used to pick yarding systems and delineate harvest units.
- Wrote R and lidR routines for individual tree crown delineation and canopy cover estimation, an open-source alternative to licensed processing software.
- Published and maintained 40+ hosted feature layers and geodatabases and managed offline map areas across three project areas. Crews used them to collect stream, inventory plot, archaeological, and forest disease and dieback data for timber sale planning, layout, and resource surveys.
- Field liaison between JRC, USFS timber staff, and National Forest Foundation partners. Supported contract administration and USFS best management practice compliance under a Registered Professional Forester.

Forester

Apr 2022 – Jul 2023 · Full-time, 50 hrs/wk

Sunrise Pines Forestry, Flagstaff, AZ (projects in seven states)

- Laid out timber sales in seven states, from Rocky Mountain mixed conifer to Gulf Coastal Plain pine. Ground-based systems throughout, cable systems on steep ground in California and Montana.

- Cruised timber by fixed-plot, variable-radius, strip, and 100% methods on varied forest types and ownerships, and ran 250+ plots myself in eastern Oklahoma. Compiled the cruise data into volume summaries for appraisals, management plans, and harvest scheduling.
- Marked timber to silvicultural prescriptions in stands from ponderosa pine and mixed conifer to bottomland hardwood and shortleaf pine. Worked out implementation with landowners, consulting foresters, and logging contractors.

Fuels Technician

Jan 2022 – Apr 2022 · Full-time, 40 hrs/wk

Fire Safe Council of Nevada County, Grass Valley, CA

- Carried out fuel reduction prescriptions under a Registered Professional Forester. Ran mechanical treatments with chainsaws, tow-chippers, tracked masticators, and hand tools for community wildfire protection. Three-month position, then moved up to the forester role at Sunrise Pines.
- Constructed control lines to burn plan specifications and worked prescribed fire ignition, holding, and mop-up operations.

Graduate Teaching Assistant

Aug 2019 – Jan 2022 · Part-time, 20 hrs/wk

Stephen F. Austin State University, Nacogdoches, TX

- Taught lab sections of 30–50 undergraduate students per semester in Introduction to Forestry and Forest Biometrics. Led field trips and taught mensuration, sampling design, and spatial analysis.
- Assisted faculty at the six-week summer field station program. Hauled and staged field equipment and kept accountability for 100+ students through exercises in timber cruising, silviculture, and forest management planning.

Fuels Technician

Nov 2017 – May 2019 · Part-time, 20 hrs/wk

TimberPro Forest Service, Nacogdoches, TX

- Operated D5 bulldozer to construct control lines and suppress slopovers and spot fires during prescribed burn operations across East Texas pine plantations.
- Ignited prescribed burns with drip torches and inspected control lines by ATV. Suppressed spot fires with hand tools across multiple burn units.

SELECTED PROJECTS

- **Mohawk Valley West Slope harvest planning (2026):** USGS 3DEP QL1 LiDAR to 3 ft DTM, CHM, and planning-slope rasters (PDAL, GDAL); 24 demonstration units, 1,621 gross ac; skyline screen of 8,540 corridors from 314 candidate landings, PNW-39 payload estimates; 25-sheet 11×17 PyQGIS map series, GeoPDFs for Avenza; simulated BAF 20 cruise checked to FSH 2409.12.
- **Forest Inventory Analyzer (Rust, v0.2.0):** cruise compiler with Survey123/Field Maps-style import, BAF expansion, TPA, basal area, QMD, diameter-class tables, Student's t sampling error and growth projections; CSV/JSON/Excel input; CI-tested releases.

EDUCATION

Master of Science in Forestry

2024

Stephen F. Austin State University

Thesis: *Consequences of Altered Burn Regimes on Plant Community Diversity in the Pineywoods of East Texas*

Honors: Xi Sigma Pi Forestry Honor Society

Thesis research and writing: Jul 2023 – Feb 2024

Bachelor of Science in Forestry

2019

Stephen F. Austin State University

Minor: Geospatial Science · Concentration: Fire Management

Relevant Coursework: Forest Biometrics, GIS and Remote Sensing, Fire Ecology, Spatial Science, Silviculture

CERTIFICATIONS

- **ELIGIBLE** - California Registered Professional Forester (RPF) — approved by the California Board of Forestry and Fire Protection to sit for the exam
- Qualified Cruiser (FSH 2409.12 ch. 60) — U.S. Forest Service, Region 3, certified 2022–2023 while cruising Region 3 sales for Sunrise Pines Forestry
- Tree Risk Assessment Qualification (TRAQ) — International Society of Arboriculture
- CPR Training — Current
- Valid California Driver's License — Clean record

PUBLICATIONS

Steinley, W. J., Oswald, B. P., Kidd, K. R., & Glasscock, J. (2024). *Influence of altered fire return intervals on plant diversity in the East Texas Pineywoods*. *Forest Research: Open Access*, 13(5), 529.